

Holographic Storage Quick Reference

Key Formulas

$$\Delta\theta_{\text{FWHM}} \approx \frac{\lambda}{2nL \cos \theta_B}$$

$$\kappa = \frac{\pi \Delta n}{\lambda \cos \theta_B}$$

$$\eta_{\text{max}} = \sin^2(\kappa L)$$

$$N_{\text{pages}} \approx \frac{\Theta_{\text{total}}}{2\Delta\theta_{\text{FWHM}}}$$

Design Workflow

1. Pick laser wavelength (λ).
2. Choose material (n , Δn).
3. Set thickness (L).
4. Choose Bragg angle (θ_B).
5. Compute $\Delta\theta_{\text{FWHM}}$.
6. Compute κ and η_{max} .
7. Set angular budget Θ_{total} .
8. Estimate max pages (N).
9. Adjust L , Δn , θ_B to balance capacity vs. efficiency.

Common Materials

Material	n	Δn range	Notes
Photopolymer	1.5–1.6	1×10^{-4} – 1×10^{-3}	High Δn , write-once
LiNbO ₃ (doped)	~ 2.2	1×10^{-5} – 1×10^{-4}	Rewritable, low Δn
BaTiO ₃	~ 2.4	1×10^{-5} – 1×10^{-4}	High sensitivity, unstable
PMMA (doped)	~ 1.49	1×10^{-5} – 1×10^{-4}	Stable, low sensitivity

Rules of Thumb

- **Thicker** = more pages, sharper selectivity.
- **Higher** Δn = stronger efficiency.
- **Shorter** λ = narrower selectivity.